

Assignment 2

P+7 (Oulipian language
modelling)

CART 498B

Nicolas R
Hernandez

Reflection on K+21 and K+7

The original task involved replacing the last word of each line with the token that had the seventh-highest probability according to GPT-2. Changing the k value from 7 to 21 had a significant impact on the resulting text. With k=7, the model still selected words that were relatively likely within the context of the sentence, producing substitutions that, while unexpected, often retained some connection to the original line. For example, replacing “winter” with “something” preserved the idea of an abstract concept but also maintained a degree of coherence.

When the value was changed to k=21, the selected tokens came from much lower probability predictions. As a result, the generated words were often less related to the poem sentences context, producing lines that felt funnier and did not make much sense. Tokens such as “dust” or “yellow” disrupted the original meaning of the poem’s overall coherence. This demonstrates that lower-probability predictions move farther away from what a human would consider logic. Increasing the x value therefore creates more variation and unpredictability, but at the cost of coherence.

To implement a P+7 technique where all nouns are replaced with their seventh-highest probability alternatives, I would first identify each noun in the poem. For each noun, I would create a prompt containing the text leading up to that noun, so that GPT-2 has the necessary context to make its prediction. I would then use the `get_kth_token_prediction_gpt2` helper function, modified to take a k value of 7, to predict the seventh-most probable next token. This token would replace the original noun in the text. This process would be repeated for every noun in the poem.

<https://github.com/NicolasH21/CART498-GenAI>